ISFusion-Attack: Instance-Scene Fusion Adversarial Attacks Against Vision Language Models in Autonomous Driving

Hongda Chen*, Changmiao Wang*, Yongquan Zhang†, Ahmed Elazab, Ming Li, Hai Wang†

***Abstract*—Vision Language Models (VLMs) have enhanced transparency and performance within Autonomous Driving (AD) systems, particularly in mixed traffic environments where autonomous vehicles must interact with human-driven vehicles, pedestrians, and cyclists. Despite their benefits, integrating VLMs into AD systems operating in such complex mixed traffic poses significant security and safety challenges. Current attack methods do not adequately address the specific needs of AD in mixed traffic. Single-modal attacks are limited because they only target isolated data streams, while cross-modal methods focus on single-instance images and fail to create effective adversarial driving scene images under the dynamics of mixed traffic. To overcome these challenges, we introduce ISFusion-Attack, a novel multi-modal adversarial framework designed to generate transferable adversarial examples for VLMs in AD, specifically targeting mixed traffic scenarios. Unlike existing methods concentrating on instance-level attacks guided by a single modality, ISFusion-Attack incorporates both instance- and scene-level multimodal information. This approach allows it to tackle the complexities inherent in real-world driving scenarios, especially the interactions between diverse agents in mixed traffic. The proposed ISFusion-Attack framework comprises three main modules. The modality enhancement module extracts and enhances visual and language features. The attention distraction module misguides VLMs by embedding misleading text into the instance and scene regions. Finally, the instance-scene fusion attack module combines enriched embeddings from both perspectives to enhance transferability. This framework is validated on both synthetic and real-world datasets, demonstrating superior performance in generating adversarial examples that address the complexities of real-world driving.**

***Index Terms*—Vision Language Models, Autonomous Driving, Adversarial Attacks, Instance-Scene Fusion.**

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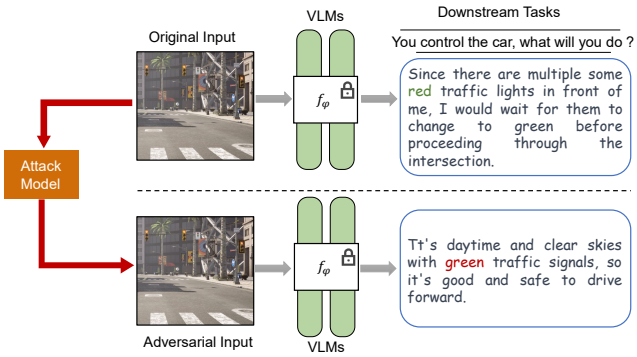


Fig. 1. Adversarial attacks pose significant risks to the accuracy and safety of VLMs in AD systems.

I. Introduction

Vision Language Models (VLMs) [1], [2], [3], [4] have been extensively studied across various domains for their remarkable abilities in generalization and interoperability. One significant area of application is Autonomous Driving (AD) [5]. During their training process, VLMs align feature embeddings from both vision and language modalities, thereby integrating the capabilities of Large Language Models (LLMs) with visual information. In the context of AD, VLM-equipped vehicles offer improved transparency in decision-making by clearly articulating the reasoning behind each decision step. This represents a significant advancement over traditional module-based and end-to-end architectures [6]. Moreover, VLMs have demonstrated promising capabilities in crucial AD tasks such as perception, prediction, planning, and control. They are particularly effective in safety-critical scenarios that involve complex road conditions and mixed traffic flows, underscoring their potential to enhance the safety and reliability of autonomous driving systems.

Despite notable achievements, VLMs still face significant security and safety concerns, especially in AD systems operating in mixed traffic. These concerns arise primarily because the adversarial vulnerabilities present in vision encoders can be transferred to VLMs, as depicted in Fig. 1. AD systems, heavily reliant on vision, are particularly susceptible to adversarial attacks on continuous and high-dimensional images captured in mixed traffic environments. It is therefore crucial to identify and understand the strategies that effectively target VLMs in AD environments, especially under the complexities of mixed traffic. This understanding is vital for recognizing

potential risks and motivating further research into advanced defense mechanisms within the community.

Traditional single-modal attack methods, such as Fast Gradient Sign Method (FGSM) [7], Iterative Fast Gradient Sign Method (I-FGSM) [8], and Projected Gradient Descent (PGD) [9], along with cross-modal attack methods like Collaborative multimodal adversarial Attack (Co-Attack) [10], Set-level Guidance Attack (SGA) [11], AttackVLM [12], and collaborative multimodal interaction attack [13], fall short of meeting the demands of AD. Single-modal attacks have limited impact since they focus on a singular stream of data, causing less disruption in the multi-modal interactions of VLMs. Existing cross-modal attacks typically concentrate on general scenarios, targeting feature space perturbations and single-instance images in digital domains. However, for real-world traffic scenes with varied road conditions and weather changes, effective attack strategies that can handle complex scene reasoning are largely unexplored. Except for the limitation of current works, another challenge in AD contexts is that reasoning tasks often allow multiple correct answers due to the complexity of the scenes. This complexity significantly increases the difficulty of crafting effective attacks. Consequently, adversarial examples must exhibit enhanced generalizability to accommodate the inherent variability in potential outcomes.

To tackle the unique attacking challenges in AD scenarios, particularly in mixed traffic conditions characterized by diverse interacting agents, we propose a robust and comprehensive framework consisting of three interconnected modules. These modules are designed to collaborate seamlessly, enabling the generation of efficient and transferable black-box adversarial examples. **(i) The Modality Enhancement (ME) module** separates vision and language data into distinct pipelines. For the vision pipeline, we utilize a pre-trained detection model to extract target instance patches. On the language side, we use Mini-GPT4 alongside semantic entropy to generate diverse, semantic-invariant captions that include reasoning about actions and perception of scenery. **(ii) The ATtention Distraction (ATD) module** focuses on misleading the attention of VLMs at both the instance and scene levels. Initially, Mini-GPT4 processes instance patches to create deceptive captions. These captions are then used to develop attention maps for the scene-level context using CLIP-based Grad-CAM, with CLIP chosen as the surrogate model due to its widespread use in VLMs. Finally, these deceptive captions are strategically embedded into the original image, guided by the generated attention maps, to effectively redirect the model's attention. **(iii) The Instance-Scene Fusion (ISF) module** leverages images from the instance and scene regions along with semantic-invariant captions to produce embeddings using multiple ensembled surrogate VLMs. These embeddings, obtained from three different levels, are collaboratively used to craft the final adversarial examples. Unlike conventional attacks on VLMs, ISF goes beyond simple vision and language interactions by incorporating instance and scene contexts, which are crucial in AD scenarios. To enhance transferability, ISF employs ensembled surrogate VLMs and enriched semantic-invariant captions.

While recent studies [14], [15] have explored similar attacks on VLMs in AD systems, our approach introduces a novel perspective with the instance and scene ensemble attack. This approach facilitates more complex interactions, significantly improving the transferability of black-box adversarial examples. Our contributions can be summarized in three key aspects:

- **Proposed ISFusion-Attack Framework:** We introduce a novel adversarial attack framework tailored for VLMs in AD scenarios. The framework integrates multiple modules to comprehensively enhance the effectiveness and transferability of adversarial examples.
- **Instance-Scene Fusion Strategy**: At the heart of our framework lies the Instance-Scene Fusion Strategy, which unifies instance context, scene context, and semantic-invariant captions within an ensembled embedding space. This approach significantly enhances the transferability and effectiveness of adversarial examples, particularly in complex autonomous driving scenarios.
- **Empirical Validation:** Extensive experiments on synthetic datasets (e.g., CARLA) and real-world benchmarks (e.g., LingoQA) validate the superiority of ISFusion-Attack. Results show that our framework significantly outperforms state-of-the-art methods in attacking multiple VLMs, ensuring consistent robustness in both synthetic and real-world AD scenarios.

The remainder of this paper is structured as follows: Section II reviews related works on adversarial attacks targeting VLMs in autonomous driving systems. Section III provides a mathematical formulation of the attack process for clarity. Section IV presents our proposed method, followed by detailed experimental results in Section V. Finally, we conclude the paper in Section VI.

II. Related Work

A. Vision-Language Models

Advancements in LLMs, exemplified by GPT-4, have catalyzed the development of VLMs, enabling sophisticated multimodal tasks by combining textual and visual understanding. These models rely on vision-language pre-training (VLP) architectures, generally categorized into single-stream and dual-stream designs.

Single-stream architectures employ a unified fusion encoder to jointly process image-text pairs, excelling in classification tasks that require integrated reasoning. Conversely, dual-stream architectures utilize separate encoders for text and image inputs, enabling efficient feature extraction and excelling in retrieval tasks where independent processing is beneficial. However, these architectures often face trade-offs in balancing speed, accuracy, and interpretability, particularly in complex applications.

VLMs have been integrated into AD system components, including perception, prediction, planning, and control [16]. They enhance system capabilities by improving explainability, facilitating human interaction, and enabling natural language-driven commands [5]. Recent studies have further explored VLMs for driver behavior recognition [17], traffic sign understanding [18], and even end-to-end driving decision-making

[19], demonstrating their expanding role in intelligent transportation systems. Despite these advantages, the relatively straightforward implementations of VLMs in AD systems expose them to adversarial vulnerabilities, necessitating robust defenses to ensure their reliability and safety in real-world, high-stakes scenarios.

B. Adversarial Attacks

Adversarial attacks involve crafting inputs designed to mislead models, exploiting their weaknesses to produce incorrect outputs. These attacks are a significant concern due to their transferability, which allows them to succeed even when applied to models in black-box settings. In autonomous driving, adversarial threats have been extensively studied across multiple modalities, including LiDAR-based object detection [20], 3D perception systems [21], and even vehicle-to-vehicle communication protocols [22]. While early studies mainly focused on classification or detection tasks [23], recent research has shifted toward generative and multimodal models, uncovering new vulnerabilities in their complex reasoning processes.

Typographic Attacks. VLMs, particularly those trained with CLIP [24], are highly susceptible to typographic attacks. These attacks manipulate visual inputs by embedding misleading text into images, disrupting zero-shot classification. For example, adversaries can digitally overlay deceptive text or physically obscure objects with carefully crafted letters. Defenses against such attacks often involve enhancing prompts or improving text-vision alignment [25]. Our work builds upon these strategies by integrating instance- and scene-level contexts, addressing the challenges of applying typographic attacks in dynamic AD environments.

Adversarial Attacks on VLMs. The integration of vision and language in VLMs introduces unique adversarial vulnerabilities. Cross-modal attacks target shared feature spaces to disrupt tasks like image-text retrieval, visual entailment, and visual grounding [26]. Early single-modal attacks, such as FGSM [7], I-FGSM [8] and PGD [9], focused on perturbing vision or language streams independently. However, these approaches often fail to exploit the intricate reasoning mechanisms inherent in VLMs, particularly in real-world applications requiring scene-level analysis.

Recent advancements, such as Co-Attack [10] and AttackVLM [12], have demonstrated efficacy in single-instance, digital scenarios but remain limited in addressing the scene-level complexities of real-world AD. Meanwhile, other studies have explored physical-world attacks, such as ghost object injection [27], LiDAR spoofing [28], and adversarial patches against traffic sign recognition [29], highlighting the diverse attack surfaces in autonomous systems. Our proposed framework addresses this gap by crafting adversarial examples that leverage both instance- and scene-level multimodal interactions, ensuring their effectiveness in diverse and dynamic environments.

Adversarial Vulnerabilities in AD. Autonomous driving systems heavily rely on vision encoders for critical tasks like perception, planning, and control [6]. This reliance makes them particularly vulnerable to adversarial perturbations, which can compromise safety and reliability. For instance, adversarial attacks can lead to misdetection of pedestrians [30], erroneous lane detection [31], or even failure in obstacle avoidance [32]. Moreover, the integration of multi-modal data introduces additional vulnerabilities, as seen in attacks on sensor fusion systems [33] and HD map-based detection [34]. While notable works [14], [15] have introduced adversarial strategies tailored to AD, they often overlook the multimodal interactions central to VLM-equipped systems. Moreover, the inherent flexibility of reasoning tasks in AD, which may permit multiple valid outcomes, complicates the creation of effective adversarial examples. This challenge is further exacerbated by the need to account for uncertainty in perception [35], distribution shifts [36], and real-world environmental variability [37], all of which are active research areas in autonomous driving safety. Our framework addresses these challenges by integrating instance- and scene-level contexts into the attack generation process. This holistic approach enables the creation of adversarial examples that are not only highly transferable across models but also adaptable to the unique demands of AD scenarios. By bridging multimodal reasoning with scene-level complexity, our work sets a new benchmark for adversarial attack methodologies in safety-critical applications.

III. Preliminaries

In Vision Question Answering (VQA) tasks within AD systems, a multimodal dataset of image-text pairs is utilized. Each sample from this dataset is represented as (v, t), where v is an image captured by a car-mounted camera and t is a caption tailored for specific tasks, such as action reasoning or scenery perception. Multiple captions can be associated with a single image, forming a set denoted as S_t. To improve the transferability of adversarial examples, the original images are adjusted to various sizes, represented as S_v. Thus, (S_v, S_t) represents a matched pair of image and caption sets.

VLMs typically comprise three primary components: an image encoder E_I, a text encoder E_T, and, in fused VLMs, a multimodal encoder E_M. VLMs generally follow either a fused or an aligned architectural pattern, with the multimodal encoder present only in the fused models. For an image v, its embedding is produced via the image encoder:

$$\boldsymbol{e}_v = E_I(v). \tag{1}$$

Similarly, a caption t is embedded by the text encoder:

$$\boldsymbol{e}_t = E_T(t). \tag{2}$$

In fused VLMs, a combined multimodal embedding is created by processing the image and text embeddings through the multimodal encoder:

$$\boldsymbol{e}_{v\&t} = E_M(\boldsymbol{e}_v, \boldsymbol{e}_t). \tag{3}$$

Additionally, the parameter ϵ_v defines the maximum allowable perturbation for images, with the permissible search space $\mathcal{B}[v, \epsilon_v]$ specified as:

$$||v^{adv} - v||_p \leq \epsilon_v, \tag{4}$$

where $||\cdot||_p$ indicates the p-norm. Similarly, the permissible alteration space for text t is denoted as $\mathcal{B}[t, \epsilon_t]$, with ϵ_t specifying the maximum number of words that can be altered.

The transition from LLMs to VLMs involves a two-stage training process, viewed from two attacking perspectives. One perspective aims to disrupt alignment through attacks on either visual or language inputs. The other targets the instruction tuning phase, introducing errors in the answer format transition. This study concentrates on the first perspective within AD systems.

When an image-text pair (v, t) is processed through an attacking model $\mathcal{A}$, an adversarial image v^{adv} is generated:

$$v^{adv} = \mathcal{A}(v, t). \tag{5}$$

As this adversarial image passes through VLMs, it can produce a targeted or non-targeted output:

$$\hat{y} = VLM(v^{adv}). \tag{6}$$

In non-targeted attacks, the goal is to maximize the semantic difference between the predicted output $\hat{y}$ and the true output y. Conversely, targeted attacks not only aim for this discrepancy but also strive to minimize the distance between $\hat{y}$ and a specific incorrect target output y_{victim}. This study aims to develop an efficient algorithm $\mathcal{A}(v, t)$ that produces a transferable adversarial image v^{adv} for non-targeted attacks in AD systems.

In summary, this section establishes the foundational concepts and notations for VQA tasks within AD systems. we introduce the multimodal dataset structure, the components of VLMs, and the embedding mechanisms for images and texts. Additionally, we delineate the constraints for adversarial perturbations in both image and text modalities, emphasizing the search spaces defined by ϵ_v and ϵ_t. Finally, we describe the objectives of both targeted and non-targeted attacks, highlighting the study's emphasis on generating transferable adversarial images in non-targeted scenarios.

IV. Methodology

In this section, we present the ISFusion-Attack framework, which includes three key modules: *Modality Enhancement*, *Attention Distraction*, and *Instance-Scene Fusion Attack*, as depicted in Fig. 2. Each component is discussed in detail in the following subsections.

A. Modality Enhancement Module

In AD systems utilizing VLMs, inputs come from both visual and linguistic sources. Our attack specifically targets the VQA tasks. As previously noted, AD tasks present distinct challenges compared to general VQA tasks. Therefore, enhancing the input modalities is crucial for effective adversarial attacks.

In the visual modality, images captured by vehicles in mixed traffic contain various objects like other cars (both autonomous and human-driven), traffic lights, motorcycles, pedestrians, and road signs. To attack a specific instance within these complex mixed traffic scenes, accurate localization is required. We employ YOLO-World [38], an open-vocabulary object detector that identifies objects beyond predefined categories.

For each driving scene image v, we target five safety-critical objects commonly encountered in mixed traffic: vehicles (including both autonomous and human-driven), traffic lights, motorcycles, pedestrians, and road signs for the attack. The set of detected instance patches for image v is expressed as:

$$\mathcal{S}_v^{instance} = \{p_i | Det(v)\}, \tag{7}$$

where $Det(\cdot)$ denotes the YOLO-World detector and p_i represents an instance patch. These detected patches narrow the attack's focus, enhancing efficiency for subsequent modules.

In the language modality, the complexity of driving scenes results in numerous possible answers for the same VQA task, complicating the crafting of effective attacks. To generate more transferable adversarial examples, we use Mini-GPT4 to produce diverse captions divided into action reasoning and scenery perception categories. To maintain semantic consistency with the original captions, we employ Semantic Entropy (SE) [39] to assess the quality of generated captions. This approach is similar to previous methods [40], but we focus on reasoning answers instead of prompts.

For each image caption t, its generated variant is t_i. To balance semantic invariance and caption diversity, the penalty function is defined as:

$$\mathcal{P}(t, t_i) = (1 - \gamma) \cdot SE(t_i) + \gamma \cdot \mathcal{D}(t, t_i), \tag{8}$$

where $\mathcal{D}(t, t_i)$ computes caption similarity using pre-trained BERT [41] embeddings and cosine similarity:

$$\mathcal{D}(t, t_i) = 1 - \frac{\phi(t) \cdot \phi(t_i)}{||\phi(t)|| \cdot ||\phi(t_i)||}, \tag{9}$$

with $\phi(\cdot)$ as the BERT embedding and γ balancing entropy reduction and semantic alignment. The enhanced caption set for t is:

$$\mathcal{S}_t = \{t_i | VLM(t, v), min\mathcal{P}(t, t_i)\}, \tag{10}$$

where $VLM(\cdot, \cdot)$ denotes Mini-GPT4, v represents the driving scene image. These enhanced image captions will aid subsequent attacks in producing more transferable adversarial examples.

In summary, we introduce the modality enhancement module, which refines both visual and linguistic inputs to enable more effective adversarial attacks in AD systems. For the visual modality, we utilize YOLO-World to detect safety-critical objects within complex driving scenes, focusing the attack on specific instance patches. In the language modality, we employ Mini-GPT4 to generate diverse captions categorized into action reasoning and scenery perception, ensuring semantic consistency through a penalty function that balances semantic alignment and diversity using semantic entropy and BERT embeddings. These enhanced modalities provide a robust foundation for crafting transferable adversarial examples, improving both attack efficiency and effectiveness.

B. Attention Distraction Module

Typographic attacks on VLMs have shown significant effectiveness [42], yet their application in complex traffic scenarios

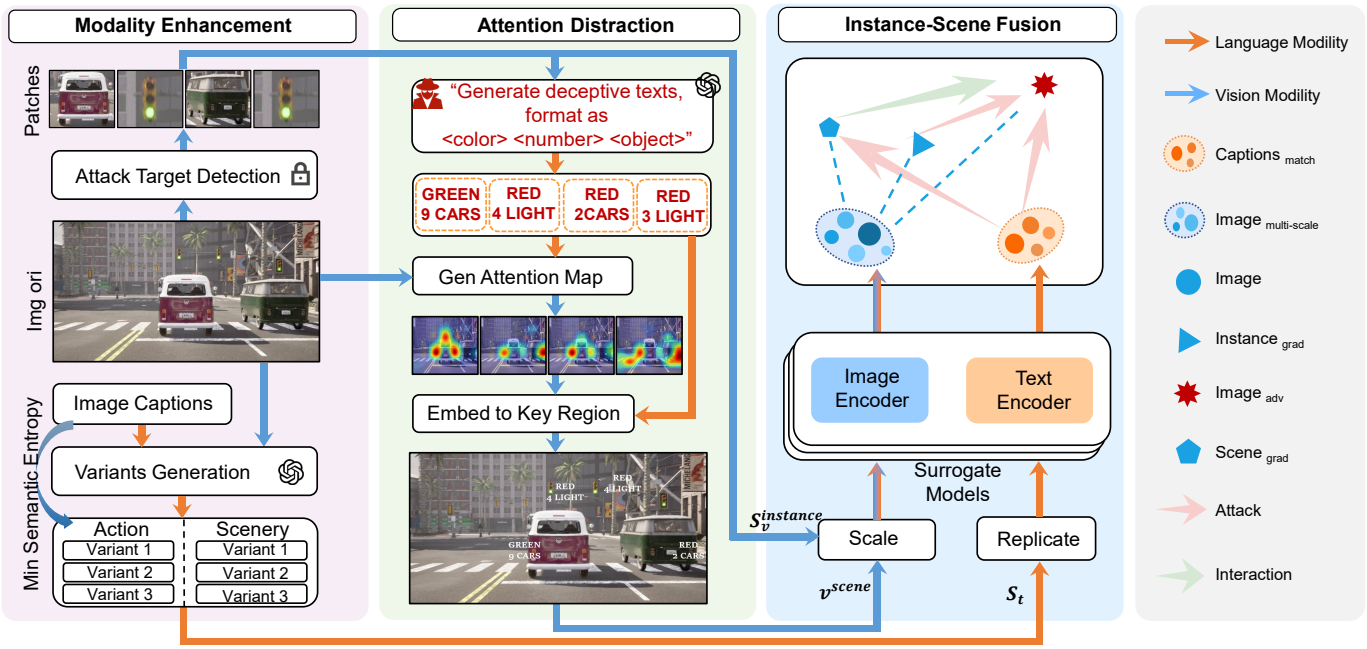


Fig. 2. Framework of the proposed ISFusion-Attack. The architecture consists of three modules: (1) The ME module separates vision and language pipelines, extracting target instance patches with a detection model and generating diverse, semantic-invariant captions using Mini-GPT4 and semantic entropy; (2) The ATD module misguides VLMs by generating deceptive captions for instance patches with Mini-GPT4, creating scene-level attention maps via CLIP-based Grad-CAM, and embedding these captions into the image to redirect the attention; (3) The ISF module generates embeddings from instance contexts, scene contexts, and captions using ensembled surrogate VLMs, crafting adversarial examples for AD scenarios while ensuring enhanced transferability through model ensembling and enriched captions.

remains underexplored. Existing typographic designs [14], [15] mainly focus on instance-level attacks in general contexts, without addressing scene-level typographic challenges in traffic settings. Recognizing this gap and inspired by the mechanisms of typographic attacks, we propose a module designed to manipulate the attention of VLMs' image encoders by integrating both instance-level and scene-level contextual information.

For instance-level contexts, deceptive captions for each instance are generated using Mini-GPT4. The content and format of these texts are crucial to the success of the attacks. In this study, we focus on images as the visual representation, emphasizing scene perception over action reasoning tasks. Specifically, we target object recognition, visual attribute detection (such as color judgment), and counting in driving scenes, which are fundamental for understanding mixed traffic. The deceptive captions are designed to be simple and concise. Thus, the instruction given to Mini-GPT4 is to "Generate deceptive texts, format as <color> <number> <object>". For the set of instance patches $\mathcal{S}_v^{instance}$ from each driving scene image v, the deceptive captions are generated as:

$$\mathcal{S}_v^{caption} = \{t_i | VLM(p_i, instruct)\}, \tag{11}$$

where t_i represents the deceptive captions for patch p_i using Mini-GPT4, with the $instruct$ is "Generate deceptive texts, format as <color> <number> <object>".

Research on typographic attacks [42] indicates that the effectiveness of these attacks is influenced by font size and opacity, with white being the most effective font color. However, there is no clear correlation between spatial positioning and attack effectiveness. Unlike previous methods [42], [14], [15], which directly overlay deceptive captions on instances, we use the attention map of deceptive texts to guide spatial positioning at the scene level.

For scene-level contexts, we hypothesize that they significantly affect the reasoning process likelihood in mixed traffic. For example, a "person" is more likely to be associated with the "crosswalk" than the "high-speed lane" in mixed traffic. Therefore, we generate a scene-level attention map using CLIP-based Grad-CAM [43]. The attention map set for a driving scene image v is:

$$\mathcal{M}_v^{scene} = \{att_i | Gradcam(v, t_i)\}, \tag{12}$$

where att_i denotes the scene-level attention map for the deceptive caption t_i, and $Gradcam$ refers to the CLIP-based Grad-CAM. We then embed the deceptive captions into their most attentive regions:

$$v^{scene} = Embed(\mathcal{M}_v^{scene}, \mathcal{S}_v^{caption}), \tag{13}$$

where $Embed(\cdot, \cdot)$ is applied to each pair of (t_i, att_i) to identify the critical regions and overlay t_i onto the original image v. This process results in a scene-level attention-guided typographic adversarial image v^{scene}.

Utilizing a scene-level attention map significantly enhances the effectiveness of typographic attacks, as demonstrated in our experiments in Section V. We attribute this improvement to the pivotal role of instance-scene interactions in shaping the inference process of VLMs.

In summary, we introduce the attention distraction module to manipulate the attention mechanisms of VLMs by integrating instance-level and scene-level contextual information through typographic attacks. At the instance level, we use

Algorithm 1 ISFusion-Attack

Input: Image-caption pair (v, t), Image encoder E_I, Text encoder E_T, Image perturbation budget ϵ_v, Iterations K, Loss function $\mathcal{L}$, Image scale set $S_s = \{s_1, ..., s_Q\}$, Number of paired captions M, Model ensemble $\mathcal{F} = \{f_i\}_{i=1}^N$, Decay factor μ, Step sizes r, β, α.

Build attack target patches $\mathcal{S}_v^{instance}$ by Eq. (7).
Build semantic-invariant captions $\mathcal{S}_t$ by Eq. (10).
Build an attacked scene image v^{scene} by Eq. (13).
// Build adversarial captions $S_{t'} = \{t_1^{'}, ..., t_M^{'}\}$.
for $i = 1, 2, \ldots, M$ **do**
  $t_i^{'} = \underset{t_i^{'} \in B[t_i, \epsilon_t](E_T, E_I) \in \mathcal{F}}{argmax} \sum - \frac{E_T(t_i^{'}) \cdot E_I(v^{scene})}{||E_T(t_i^{'})|| \cdot ||E_I(v^{scene})||}$.
end for
// Build an instance-scene image set,
// $S_v^{ins-sce} = \{(v_i^{ins}, v_i^{sce}), ..., (v_Q^{ins}, v_Q^{sce})\}$.
for $i = 0, 1, \ldots, Q-1$ **do**
  $v_i^{ins} = resize(p_i, s_i) + 0.05 \cdot \mathcal{N}(0, 1)$.
  $v_i^{sce} = resize(v^{scene}, s_i) + 0.05 \cdot \mathcal{N}(0, 1)$.
end for
Initialize: Image set $S_{v_0}^{ins-sce} = S_v^{ins-sce}$,
outer momentum $\mathbf{m} = 0$, inner momentum $\hat{\mathbf{m}} = 0$.
for $k = 0, 1, \ldots, K-1$ **do**
  Calculate $\mathbf{g} = \nabla_v \mathcal{L}(\sum_{i=1}^N f_i(S_{t'}, S_{v_k}^{ins-sce}))$.
  Update v_k by $v_k^0 = clip_{v,\epsilon_v}(v_k + r \cdot sign(\mathbf{g}))$.
  for $i = 1, 2, \ldots, N$ **do**
    Calculate $\mathbf{g} = \nabla_v \mathcal{L}(\sum_{i=1}^N f_i(S_{t'}, S_{v_k^{i-1}}^{ins-sce}))$.
    Update inner momentum by $\hat{\mathbf{m}} = \mu \cdot \hat{\mathbf{m}} + \frac{\mathbf{g}}{||\mathbf{g}||_2}$.
    Update v_k^i by $v_k^i = clip_{v,\epsilon_v}(v_k^{i-1} - \beta \cdot \hat{\mathbf{m}})$.
  end for
  Calculate the update $\mathbf{g} = v_k^N - v_k$.
  Update momentum $\mathbf{m} = \mu \cdot \mathbf{m} + \mathbf{g}$.
  Update v_{k+1} by $v_{k+1} = clip_{v,\epsilon_v}(v_k + \alpha \cdot sign(\mathbf{m}))$.
end for

Output: The final adversarial image $v^{adv} = v_K$.

Mini-GPT4 to generate concise, deceptive captions for key image patches, guiding the attack focus. At the scene level, we apply CLIP-based Grad-CAM to generate attention maps that highlight regions most relevant to the deceptive captions, enabling precise embedding of typographic modifications. This dual-layer approach allows us to exploit both object-specific and contextual scene information, effectively addressing the unique challenges of traffic scenarios and significantly improving the performance of typographic adversarial attacks.

C. Instance-Scene Fusion Attack

Inspired by recent developments [44], [11], we incorporate the concept of instance-scene fusion into cross-modal interactions and employ a transferability-friendly model ensemble strategy. Through the first two modules, we compile the attack instance patches $\mathcal{S}_v^{instance}$, semantic-invariant captions $\mathcal{S}_t$, and a typographically attacked scene image v^{scene}. Using these components, we aim to craft a robust adversarial example.

Traditional methods often rely solely on instance contextual information to create adversarial examples. In contrast, we leverage augmented instance-scene contextual information, which includes a set-level integration of images and captions. From the diverse semantic-invariant captions $\mathcal{S}_t$ and scene image v^{scene}, we construct an adversarial caption set $\mathcal{S}_{t'} = \{t_1^{'}, \cdots, t_M^{'}\}$, where M is the number of paired captions. To enhance the robustness of adversarial texts, we ensemble more surrogate models $\mathcal{F} = \{f_i\}_{i=1}^N$. Each $t_i^{'}$ is constructed as follows:

$$t_i^{'} = \underset{t_i^{'} \in \mathcal{B}[t_i, \epsilon_t](E_T, E_I) \in \mathcal{F}}{argmax} \sum - \frac{E_T(t_i^{'}) \cdot E_I(v^{scene})}{||E_T(t_i^{'})|| \cdot ||E_I(v^{scene})||}. \quad (14)$$

The goal is to minimize the distance between the adversarial caption $t_i^{'}$ and the scene image v^{scene} within the embedding space.

To enhance interaction between instance and scene contexts, we further augment the visual modality by resizing all images to various scales $\mathcal{S}_s = \{s_1, ..., s_Q\}$ and adding Gaussian noise, creating a multi-scale instance-scene image set:

$$\mathcal{S}_v^{isc} = \{(v_i^{ins}, v_i^{sce}), ..., (v_Q^{ins}, v_Q^{sce})\}, \quad (15)$$

where Q represents the number of image scales. Each pair (v_i^{ins}, v_i^{sce}) is formulated as:

$$v_i^{ins} = resize(p_i, s_i) + 0.05 \cdot \mathcal{N}(0, 1), \quad (16)$$

$$v_i^{sce} = resize(v^{scene}, s_i) + 0.05 \cdot \mathcal{N}(0, 1), \quad (17)$$

where p_i is an instance patch from Eq. (7) and v^{scene} is the scene image from Eq. (13).

These augmented modalities are then employed in the embedding space to execute cross-modal attacks. Specifically, we utilize the enriched instance and scene contexts to interact within an ensemble embedding space. The combined information serves as guidance, directing the optimization process for generating adversarial data.

For adversarial image generation, we define the following loss function:

$$\mathcal{L} = - \sum_{t_i^{'} \in \mathcal{S}_{t'}} \frac{E_T(t_i^{'})}{||E_T(t_i^{'})||} \cdot \sum \frac{cat(E_I(v_i^{ins}), E_I(v_i^{sec}))}{||cat(E_I(v_i^{ins}), E_I(v_i^{sec}))||}, \quad (18)$$

where $(v_i^{ins}, v_i^{sec}) \in \mathcal{S}_v^{isc}$ and cat$(\cdot, \cdot)$ denotes the concatenation of the instance and scene context information. The final objective is:

$$v^{adv} = \underset{v^{adv}}{argmax} \mathcal{L}(\sum_{i=1}^N f_i(\mathcal{S}_{v^{adv}}^{isc}, \mathcal{S}_{t'})), s.t. ||v^{adv} - v||_\infty \leq \epsilon_v, \quad (19)$$

where we minimize the distance between multi-scale instance-scene contextual images and adversarial captions in the ensemble embedding space, while ensuring the ℓ_∞ distance between v and v^{adv} is less than ϵ_v.

To solve the optimization in Eq. (19), we adopt advanced transfer-based attack methods [44]. The Spectrum Simulation Attack (SSA) [45] enhances adversarial transferability via spectrum transformations. Similarly, the Common Weakness Attack (CWA) [44] identifies vulnerabilities across an ensemble of surrogate models by promoting loss landscape flatness and aligning local optima. By integrating SSA and CWA, the

combined SSA-CWA method achieves superior transferability in black-box settings. A summary of our framework is detailed in Algorithm 1.

In summary, we introduce a fusion module that combines instance and scene information to generate cross-modal adversarial examples. This module integrates semantic-invariant captions with typographically altered scene images. By incorporating both instance-level and scene-level contextual data, we enhance the robustness of our approach through multi-scale augmentation and the addition of Gaussian noise. The adversarial examples are refined within an ensemble embedding space using an innovative loss function. Furthermore, we implement sophisticated techniques such as SSA and CWA to enhance the transferability of these adversarial examples in black-box environments. Our framework effectively tackles the challenges posed by driving scenarios, especially mixed traffic, by leveraging rich multi-modal contexts to create robust cross-modal attacks.

V. Experiments

The comprehensive pipeline illustrated in Fig. 3 outlines the entire experimental process. Initially, the original input is processed by the attacking models to create adversarial images. Following this, VLMs are employed to analyze both the original and adversarial images, generating answers to specific questions posed for each scenario. The effectiveness of the attacking models is then evaluated using four distinct metrics. Detailed explanation of each stage within this pipeline is provided in the subsequent subsections.

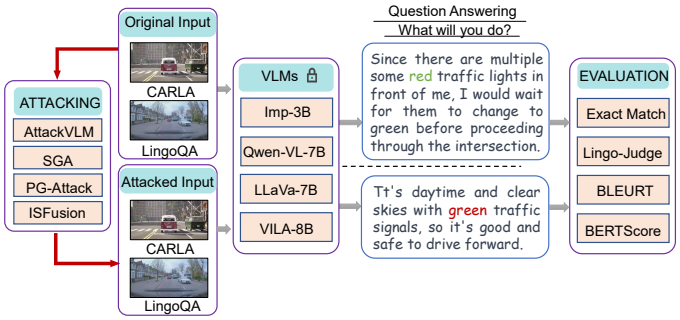


Fig. 3. A comprehensive experimental pipeline. Firstly, the original input goes through the attacking models to output the adversarial images. Secondly, VLMs infer original input and adversarial input to get the answers to specific questions, respectively. Lastly, both the answers are input into the evaluation metric to get the scores of these attacking models.

A. Experimental Setup

Dataset. In our experiments, we employ VLMs on VQA datasets tailored for AD systems, specifically utilizing LingoQA [5] and the CARLA [14] dataset. LingoQA is divided into three components: the scenery dataset, which tests description and localization abilities; the action dataset, focusing on action justification skills; and the evaluation dataset, which encompasses all of these competencies. For our experiments, we exclusively use the evaluation dataset from LingoQA, testing over 1000 question-answer pairs across 100 images. Each image is accompanied by at least five questions concerning action, justification, counting, and localization.

Regarding the CARLA dataset, it originates from the CARLA simulator and includes scenarios simulating mixed traffic conditions. Phase 1 comprises 500 images with a relatively balanced object distribution, whereas Phase 2 contains 100 images with a highly unbalanced object distribution, better representing challenging mixed traffic situations. This subset consists of over 200 question-answer pairs based on 100 images, concentrating on scene reasoning, such as target counting and instance perception across five classes essential for mixed traffic: cars (autonomous and human-driven), persons, motorcycles, traffic lights, and road signals.

Baseline Methods. To evaluate the performance improvement of ISFusion-Attack, we compare it against several established baselines: AttackVLM [12], SGA [11], and PG-Attack [14]. AttackVLM offers a transferable targeted attacking strategy that supports query-based attacks, enhancing robustness in black-box environments. However, it is primarily suited for digital-world images with single-object scenes. SGA enhances Co-Attack by incorporating set-level alignments to improve the transferability of adversarial examples. PG-Attack builds on CMI-Attack and is specifically designed for autonomous driving systems. In our experiments, we employ specific surrogate models for transfer-based methods. Consistent with previous setups, Mini-GPT4 is used as the surrogate model for AttackVLM, while ALBEF, TCL, and VIT-B-16 are employed for SGA.

Evaluation Metrics. Exact Match (EM), Lingo-Judge Accuracy (LJ), BLEURT (BU), and BERTScore (BT) can be used as metrics to assess the similarity between the output of the attacked VLM and the ground truth text. Lower values of the metrics indicate a higher Attack Success Rate (ASR). To facilitate the data analysis of the experimental results, we define the attack success rate as $\mathcal{ASR}_{sim} = 1 - Score_{sim}$ based on these similarity metrics, where $Score_{sim}$ denotes the similarity derived from the EM, LJ, BU or BT Score. With this setting, a higher value of $\mathcal{ASR}_{sim}$ indicates that the attack method is more effective.

Implementation Details. For the modality enhancement module, we target five classes: cars, motorcycles, traffic lights, road signals, and persons. To balance semantic alignment and entropy reduction, we set γ in Eq. (8) to 0.5 and use 30 paired captions ($M = 30$). The image scale range is defined as $S_s = \{0.50, 0.75, 1.00, 1.25, 1.50\}$. Surrogate models such as Mini-GPT4, CLIP, and BLIP-2 are used to encode both visual and language data. Following prior work [11], the perturbation budget is set at $\epsilon_v = 2/255$ under the ℓ_∞ norm and $\epsilon_t = 1$. For the training loop, with iteration steps $K = 10$, the step size is set to $\alpha = 0.5/255$, $\beta = 600$, and $r = 1/255$. The VLMs targeted for attacks included LLaVa [1], VILA [2], Qwen-VL [3], and Imp [4]. All experiments are conducted on an NVIDIA A800 GPU with 80 GiB of memory.

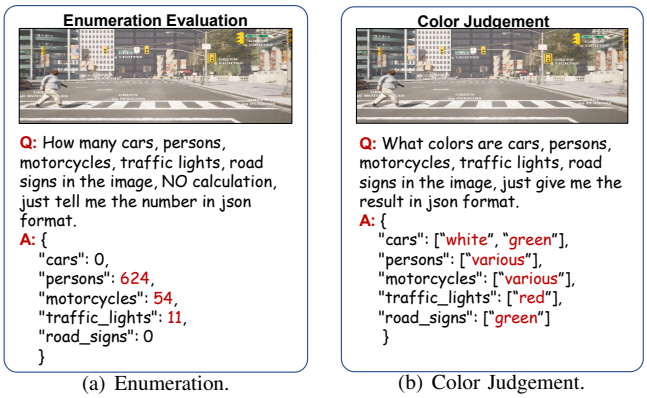
B. Attacking Performance

As shown in Table I, our proposed ISFusion-Attack framework achieves consistently higher attack success rates ($\mathcal{ASR}_{sim}$) across all evaluation metrics compared to baseline methods. The table compares various attack strategies

TABLE I

COMPARISON OF DIFFERENT ATTACK STRATEGIES ON THE CARLA AND LINGOQA DATASETS. $\mathcal{ASR}_{EM}$, $\mathcal{ASR}_{LJ}$, $\mathcal{ASR}_{BU}$, AND $\mathcal{ASR}_{BT}$ DENOTE THE ATTACK SUCCESS RATES MEASURED BY EXACT MATCH, LINGO-JUDGE ACCURACY, BLEURT, AND BERTSCORE, RESPECTIVELY. HIGHER VALUES INDICATE MORE EFFECTIVE ATTACKS. THE **IMPROVEMENT** ROW HIGHLIGHTS THE PERFORMANCE GAINS OF OUR METHOD COMPARED TO THE BEST-PERFORMING BASELINE.

Target Model	Attack Method	CARLA [14] (object recognition and counting)				LingoQA [5] (scenery and action reasoning)			
		ASR_{EM}(↑)	ASR_{LJ}(↑)	ASR_{BU}(↑)	ASR_{BT}(↑)	ASR_{EM}(↑)	ASR_{LJ}(↑)	ASR_{BU}(↑)	ASR_{BT}(↑)
Qwen-VL-7B	*attackVLM/miniGPT4* [12]	0.650	0.510	0.110	0.042	0.819	0.661	0.605	0.131
	SGA/ALBEF [11]	0.720	0.580	0.211	0.025	0.850	0.636	0.596	0.129
	SGA/TCL [11]	0.726	0.565	0.207	0.044	0.870	0.638	0.595	0.129
	SGA/VIT-B-16 [11]	0.740	0.565	0.210	0.044	0.819	0.640	0.598	0.130
	PG-Attack [14]	0.745	0.585	0.212	0.045	0.980	0.654	0.568	0.127
	ISFusion-Attack (Ours)	**0.980**	**0.770**	**0.331**	**0.074**	**0.999**	**0.973**	**0.805**	**0.181**
	Improvement	0.235↑	0.185↑	0.119↑	0.029↑	0.019↑	0.312↑	0.200↑	0.050↑
Imp-3B	*attackVLM/miniGPT4* [12]	0.599	0.475	0.129	0.015	0.969	0.666	0.586	0.121
	SGA/ALBEF [11]	0.704	0.624	0.149	0.019	0.968	0.668	0.589	0.120
	SGA/TCL [11]	0.690	0.610	0.151	0.018	0.975	0.648	0.583	0.119
	SGA/VIT-B-16 [11]	0.715	0.610	0.156	0.013	0.972	0.680	0.588	0.120
	PG-Attack [14]	0.740	0.635	0.158	0.020	0.978	0.657	0.497	0.111
	ISFusion-Attack (Ours)	**0.970**	**0.845**	**0.245**	**0.035**	**0.998**	**0.904**	**0.789**	**0.176**
	Improvement	0.230↑	0.210↑	0.087↑	0.015↑	0.020↑	0.247↑	0.200↑	0.055↑
VILA-8B	*attackVLM/miniGPT4* [12]	0.951	0.770	0.230	0.040	0.973	0.639	0.563	0.125
	SGA/ALBEF [11]	0.955	0.755	0.239	0.042	0.969	0.636	0.559	0.123
	SGA/TCL [11]	0.955	0.755	0.237	0.042	0.958	0.645	0.559	0.122
	SGA/VIT-B-16 [11]	0.950	0.749	0.241	0.044	0.980	0.653	0.560	0.118
	PG-Attack [14]	0.960	0.780	0.244	**0.046**	0.991	0.607	0.538	0.110
	ISFusion-Attack (Ours)	**0.985**	**0.810**	**0.359**	**0.046**	**0.999**	**0.891**	**0.764**	**0.130**
	Improvement	0.025↑	0.030↑	0.115↑	0	0.008↑	0.238↑	0.201↑	0.005↑
LLaVa-7B	*attackVLM/miniGPT4* [12]	0.435	0.350	0.115	0.015	0.974	0.631	0.574	0.127
	SGA/ALBEF [11]	0.549	0.418	0.137	0.012	0.968	0.666	0.574	0.123
	SGA/TCL [11]	0.560	0.417	0.132	0.017	0.954	0.656	0.575	0.123
	SGA/VIT-B-16 [11]	0.555	0.400	0.136	0.018	0.946	0.662	0.572	0.124
	PG-Attack [14]	0.580	0.425	0.140	0.018	0.989	0.601	0.479	0.099
	ISFusion-Attack (Ours)	**0.865**	**0.635**	**0.210**	**0.029**	**0.993**	**0.908**	**0.776**	**0.159**
	Improvement	0.285↑	0.210↑	0.070↑	0.011↑	0.004↑	0.242↑	0.201↑	0.032↑



(a) Enumeration. (b) Color Judgement.

Fig. 4. The demonstrations showcase enumeration and color judgment attacks on the CARLA dataset, targeting the Qwen-VL-7B model.

applied to digital-world datasets (CARLA [14]) and real-world datasets (LingoQA [5]), emphasizing that higher $\mathcal{ASR}_{sim}$ values reflect more effective attacks. Additionally, the improvement row highlights the significant gains achieved by our method over the next-best performing strategies across all evaluated VLMs.

Specifically, on the digital-world CARLA dataset, which focuses on object recognition and counting tasks, ISFusion-Attack outperforms competing methods by substantial margins. For Qwen-VL-7B and Imp-3B, we observe $\mathcal{ASR}_{EM}$ improvements of 23.5% and 23.0%, respectively, over the second-best methods. Similarly, $\mathcal{ASR}_{LJ}$ for Imp-3B improves by 21.0%, reflecting the effectiveness of our instance-scene fusion strategy in exploiting contextual dependencies. Notable gains in $\mathcal{ASR}_{BU}$ and $\mathcal{ASR}_{BT}$ further underline the versatility of our method across different similarity metrics.

For the real-world LingoQA dataset, which evaluates reasoning about scenery and actions, ISFusion-Attack demonstrates remarkable improvements. The method achieves $\mathcal{ASR}_{EM}$ gains of 24.2% for LLaVa-7B and 31.2% for Qwen-VL-7B compared to their respective baselines, highlighting its ability to disrupt precise matching with ground truth. $\mathcal{ASR}_{LJ}$ and $\mathcal{ASR}_{BU}$ also show significant enhancements across all VLMs, with improvements exceeding 20% in most cases. Although the improvements in $\mathcal{ASR}_{BT}$ are less pronounced, the consistent positive trend suggests that our method is capable of introducing meaningful semantic discrepancies.

Overall, the analysis highlights that ISFusion-Attack is particularly effective in scenarios requiring complex scene-context integration, as evidenced by its superior performance on real-world datasets involving mixed traffic. The method's ability to achieve higher $\mathcal{ASR}_{sim}$ across diverse metrics underscores its robustness and generalizability across different tasks and evaluation VLMs, especially in mixed traffic environments.

Intuitively, our approach demonstrates outstanding performance on both the CARLA and LingoQA datasets, as illustrated in Fig. 4 and Fig. 5, respectively. We carefully select challenging mixed traffic scenarios to evaluate the effectiveness of our ISFusion-Attack framework, including

TABLE II
ABLATION EXPERIMENTAL RESULTS OF DIFFERENT MODULES ON CARLA AND LINGOQA DATASETS. $\mathcal{ASR}_{EM}$, $\mathcal{ASR}_{LJ}$, $\mathcal{ASR}_{BU}$, AND $\mathcal{ASR}_{BT}$ DENOTE THE ATTACK SUCCESS RATES MEASURED BY EXACT MATCH, LINGO-JUDGE ACCURACY, BLEURT, AND BERTSCORE, RESPECTIVELY. HIGHER VALUES INDICATE MORE EFFECTIVE ATTACKS, WITH ME DENOTING THE MODALITY ENHANCEMENT MODULE, ATD FOR THE ATTENTION DISTRACTION MODULE, AND ISF FOR THE INSTANCE-SCENE FUSION MODULE.

Target Model	Attack Modules			CARLA [14] (Object Recognition and Counting)				LingoQA [5] (Scenery and Action Reasoning)			
	ME	*ATD*	*ISF*	ASR_{EM}(↑)	ASR_{LJ}(↑)	ASR_{BU}(↑)	ASR_{BT}(↑)	ASR_{EM}(↑)	ASR_{LJ}(↑)	ASR_{BU}(↑)	ASR_{BT}(↑)
Qwen-VL-7B		✔	✔	0.970	0.755	0.330	0.061	0.979	0.657	0.498	0.097
	✔		✔	0.946	0.686	0.228	0.045	0.981	0.622	0.255	0.085
	✔	✔		0.913	0.560	0.206	0.031	0.968	0.488	0.317	0.059
	✔	✔	✔	**0.980**	**0.770**	**0.331**	**0.074**	**0.999**	**0.973**	**0.805**	**0.181**
Imp-3B		✔	✔	0.965	0.677	0.227	0.032	0.982	0.599	0.353	0.074
	✔		✔	0.892	0.596	0.169	0.023	0.972	0.587	0.347	0.068
	✔	✔		0.897	0.585	0.148	0.018	0.955	0.511	0.279	0.043
	✔	✔	✔	**0.970**	**0.845**	**0.245**	**0.035**	**0.998**	**0.904**	**0.789**	**0.176**
VILA-8B		✔	✔	0.972	0.781	0.209	0.041	0.983	0.633	0.476	0.087
	✔		✔	0.958	0.651	0.204	0.017	0.985	0.658	0.369	0.102
	✔	✔		0.931	0.640	0.133	0.003	0.963	0.544	0.298	0.075
	✔	✔	✔	**0.985**	**0.810**	**0.359**	**0.046**	**0.999**	**0.891**	**0.764**	**0.130**
LLaVa-7B		✔	✔	0.857	0.617	0.197	0.021	0.980	0.792	0.679	0.110
	✔		✔	0.803	0.545	0.019	0.014	0.969	0.563	0.307	0.079
	✔	✔		0.844	0.547	0.091	0.017	0.947	0.466	0.221	0.037
	✔	✔	✔	**0.865**	**0.635**	**0.210**	**0.029**	**0.993**	**0.908**	**0.776**	**0.159**

complex driving scenes with intricate object interactions and adverse weather conditions such as foggy environments. These scenarios highlight the robustness and adaptability of our method in tackling real-world mixed traffic challenges.

C. Ablation Study

We perform ablation experiments to evaluate the contributions of the ME, ATD, and ISF modules to the overall performance of our attack framework. The experiments assess the performance of the full framework against its variants, where one module is removed at a time, using the CARLA and LingoQA datasets. The metrics $\mathcal{ASR}_{EM}$, $\mathcal{ASR}_{LJ}$, $\mathcal{ASR}_{BU}$, and $\mathcal{ASR}_{BT}$ provide quantitative evaluations. The results in Table II demonstrate that removing any of these modules leads to a significant decline in attack success rates, underlining their critical roles.

Performance after removing the ME module. The ME module significantly enhances the attack by enriching input features and capturing multimodal semantics. Without this module, adversarial examples fail to fully exploit modality-specific vulnerabilities, resulting in a marked decline in effectiveness.

For instance, in the CARLA dataset, removing the ME module leads to a reduction in $\mathcal{ASR}_{EM}$ for Qwen-VL-7B from 0.980 to 0.913 and in $\mathcal{ASR}_{LJ}$ from 0.770 to 0.560. Similarly, in the LingoQA dataset, $\mathcal{ASR}_{EM}$ decreases from 0.999 to 0.968, while $\mathcal{ASR}_{BT}$ drops from 0.181 to 0.059. For Imp-3B, $\mathcal{ASR}_{EM}$ and $\mathcal{ASR}_{LJ}$ decrease by 8.1% and 25.6%, respectively, highlighting the ME module's impact on object recognition and reasoning tasks.

Performance after removing the ATD module. The ATD module is essential for disrupting the VLM's attention mechanisms at both instance and scene levels. Without it, the adversarial examples are less effective at manipulating attention pathways, leading to a noticeable decline in performance metrics.

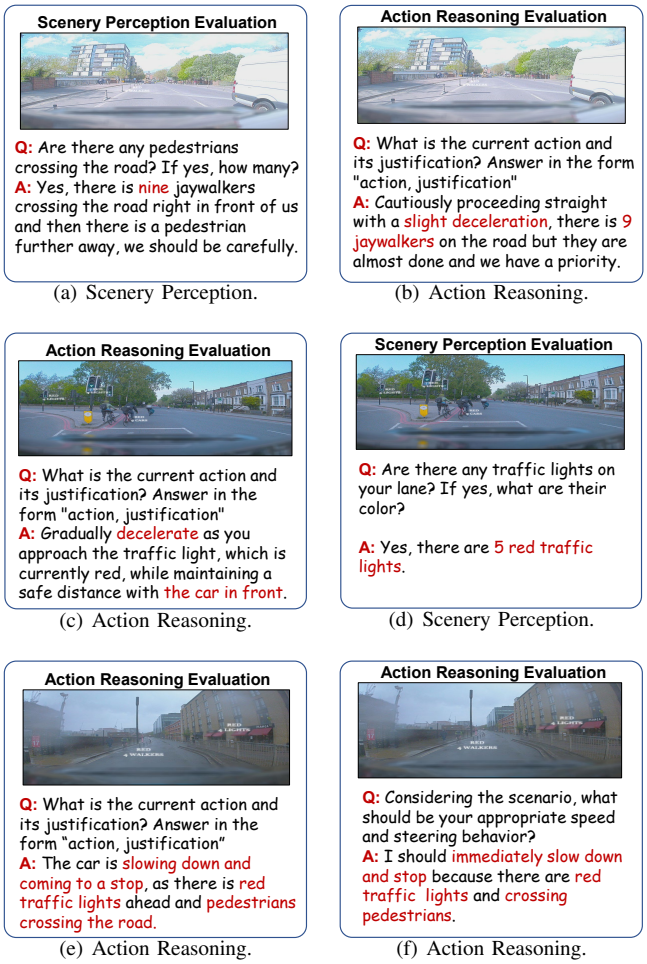
In the CARLA dataset, Qwen-VL-7B sees its $\mathcal{ASR}_{LJ}$ drop from 0.770 to 0.686 when the ATD module is removed, and $\mathcal{ASR}_{BT}$ declines from 0.074 to 0.045. Similarly, for the LingoQA dataset, $\mathcal{ASR}_{BU}$ decreases from 0.805 to 0.317. The Imp-3B model exhibits a similar trend, where $\mathcal{ASR}_{EM}$ decreases by 7.5% and $\mathcal{ASR}_{LJ}$ by 17.2%. These results emphasize the importance of ATD in redirecting the model's focus and improving adversarial attack efficacy.

Performance after removing the ISF module. The ISF module serves as the foundation of our framework, integrating instance-level, scene-level, and semantic-invariant strategies. Its absence results in incomplete adversarial examples that fail to disrupt the interactions between modalities effectively.

For Qwen-VL-7B, removing the ISF module causes a drop in $\mathcal{ASR}_{EM}$ from 0.980 to 0.970 in the CARLA dataset, and $\mathcal{ASR}_{BT}$ decreases from 0.074 to 0.061. The degradation is even more pronounced in the LingoQA dataset, where $\mathcal{ASR}_{EM}$ and $\mathcal{ASR}_{LJ}$ decline by 3.1% and 32.4%, respectively. The VILA-8B model experiences similar impacts, with $\mathcal{ASR}_{BU}$ decreasing from 0.359 to 0.209 in the CARLA dataset and from 0.764 to 0.476 in LingoQA. These results confirm the ISF module's essential role in integrating information for robust and transferable attacks.

Comparison of ME, ATD, and ISF Modules. To further understand the relative contributions of the ME, ATD, and ISF modules, we compare their individual effects on attack performance. Table II reveals that while all three modules are essential, their impacts vary depending on the dataset, model, and evaluation metric.

The **ISF module** consistently demonstrates the most substantial contribution to attack performance across both



(a) Scenery Perception. (b) Action Reasoning. (c) Action Reasoning. (d) Scenery Perception. (e) Action Reasoning. (f) Action Reasoning.

Fig. 5. The demonstrations highlight the evaluation of scenery perception and action reasoning under diverse weather conditions and complex driving scenes on the LingoQA dataset, showcasing the attacking performance of our approach against the Qwen-VL-7B model.

datasets. For instance, in the CARLA dataset with the Qwen-VL-7B model, the $\mathcal{ASR}_{EM}$ drops by 6.7% (from 0.980 to 0.913) when ISF is removed, compared to 3.5% (0.980 to 0.946) for ME and 7.1% (0.980 to 0.913) for ATD. Similarly, in the LingoQA dataset, the absence of ISF leads to a significant decline in $\mathcal{ASR}_{LJ}$ from 0.973 to 0.488, a reduction of over 49.8%, compared to reductions of 36.1% and 35.5% for ME and ATD, respectively. This emphasizes the ISF module's importance in ensuring a comprehensive fusion of instance-level and scene-level information.

The **ME module**, while slightly less critical than ISF, plays a vital role in enhancing modality-specific information. For example, in the CARLA dataset, removing ME causes a greater decline in $\mathcal{ASR}_{BT}$ (from 0.074 to 0.045) than removing ATD (from 0.074 to 0.061) or ISF (from 0.074 to 0.061) for Qwen-VL-7B. Similarly, in the LingoQA dataset, ME contributes substantially to metrics like $\mathcal{ASR}_{BU}$ and $\mathcal{ASR}_{BT}$, with decreases of 20.5% and 25.2%, respectively, when removed. This highlights ME's role in enriching adversarial examples with multimodal semantics.

The **ATD module**, while crucial for disrupting attention mechanisms, generally has a slightly smaller impact compared to ISF and ME. For instance, in the CARLA dataset, removing ATD results in a 13.7% drop in $\mathcal{ASR}_{LJ}$ for VILA-8B (from 0.810 to 0.651), compared to a 16.5% drop when ISF is removed. Similarly, in the LingoQA dataset, ATD's removal causes a decline in $\mathcal{ASR}_{EM}$ from 0.999 to 0.968 for Qwen-VL-7B, a smaller reduction than the 50.5% observed when ISF is removed. Nonetheless, ATD is indispensable for effectively manipulating attention pathways and achieving optimal attack performance.

Conclusion of Module Comparison. While the ISF module consistently provides the most significant boost in attack success rates due to its holistic integration of information across modalities, the ME module is particularly impactful for enriching input features, and the ATD module excels in attention manipulation. Together, these modules form a complementary and interdependent framework, enabling the generation of highly effective adversarial examples for both object recognition and reasoning tasks.

VI. Conclusion

In this paper, we propose a novel attack framework to address significant security challenges associated with deploying VLMs in AD systems, focusing specifically on adversarial vulnerabilities. The framework integrates three key modules, each designed to enhance attack effectiveness and address limitations of existing single-modal and cross-modal methods. The ME module introduces semantic-invariant captions and targeted instance patches to enrich adversarial inputs with diversity and contextual relevance. The ATD module leverages scene-level attention maps to embed deceptive captions, effectively misdirecting the VLM's focus. The ISF module bridges instance and scene contexts using ensemble surrogate models, generating adversarial examples with superior generalization. Extensive experiments validate the framework's ability to craft highly transferable black-box attacks, outperforming existing methods in complex and dynamic driving environments. Future work will explore potential extensions of this approach to further strengthen the security and robustness of vision-language model architectures operating in dense and interactive mixed traffic.

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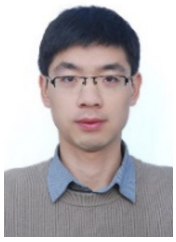

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